

B Kalyaanakrishnan

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EDUCATION

PES University | B.Tech in Computer Science and Engineering, CGPA: 8.27 Aug 2022 - Aug 2026 | Bengaluru, KA

CERTIFICATIONS

CCE, IISc Bengaluru | Large Language Models: A Hands-on Approach Jan 2026 - May 2026

EXPERIENCE

Backend Engineering Intern | Toddle - Your Teaching Partner Jan 2026 - Present | Bengaluru, KA

- Improved an LLM-based assignment-evaluation feature by engineering the regeneration prompt to include the model's prior evaluation as conversation-history context, adding a new prompt-template input variable and retrieval query against the AI conversation/message store.
- Fixed a data-integrity bug where AI-generated evaluation scores were silently dropped from the gradebook aggregated view, moving evaluation-presence checks into per-row LEFT JOINS across 3 query functions so AI ratings surfaced correctly at scale.
- Built backend platform features using GraphQL APIs, schema evolution, and resolver implementation for integration-focused workflows; hardened resolver behavior around API edge cases and failed integration states.

AI/ML Intern | Pricetime Fintech Private Limited Jul 2025 - Aug 2025 | Bengaluru, KA

- Designed and implemented a custom Model Context Protocol server connecting Claude Desktop to a local bond-market database for natural-language querying of structured financial data.
- Built database retrieval and validation workflows for bond data access, including query execution, result inspection, integration debugging, and faster developer iteration.

PROJECTS

Context Forge | Python, LLM Evaluation, LoCoMo, LongMemEval Apr 2026 - May 2026

- Built a hybrid-memory system for long-context assistants using raw evidence memory and compressed summary memory, scaling from a 1,095-chunk / 272-summary LoCoMo prototype to a 1,913-chunk / 948-summary LongMemEval pipeline with GPU-based embedding retrieval.
- Implemented and compared 3 memory-routing strategies - rule-based baseline, Wide & Deep controller, and MLP controller - across a 200-query evaluation subset; final MLP showed non-trivial routing behavior with 160 raw / 40 summary decisions.

Fake Audio Detection System | Python, TensorFlow, Scikit-learn, Flask Mar 2025 - Apr 2025

- Built FADE, a real-time deepfake-audio detection system using a 3-model ensemble across 10-second, 2-second, and short-snippet audio analysis with spectral, ZCR, RMS, MFCC, and Chroma features.
- Integrated batch-streaming prediction for live and recorded audio classification, achieving 92% overall accuracy while using a conservative flag-if-any ensemble strategy to reduce false negatives.

Distributed Task Queue | Python, Kafka, Redis Nov 2024 - Dec 2024

- Designed a distributed task queue with Python workers, Kafka producer-consumer messaging, Redis-backed task state, retries, and asynchronous execution patterns.
- Configured a 5,000-task workload across a 10-partition Kafka topic with 10 workers, adding Redis task-status tracking plus worker heartbeats with 5-second refreshes and 10-second TTLs.

Collaborative Markdown Editor | Java, Spring Boot, JavaFX, PostgreSQL Feb 2025 - Mar 2025

- Architected a Spring Boot + JavaFX collaborative editor using 5 design patterns: Strategy for pluggable CommonMark/Flexmark rendering, Observer for WebSocket/STOMP real-time sync, Command for per-document undo/redo history, Factory for PDF/DOCX/HTML export, and Repository for JPA data access.
- Optimized collaborator queries with JPQL LEFT JOIN FETCH to eliminate N+1 query issues, and implemented snapshot-based document versioning with owner/collaborator access control enforced at the service layer.

Mini Kubernetes Simulation System | Python, Docker, JavaScript Nov 2024 - Dec 2024

- Built a Kubernetes-style cluster simulation with Python orchestration, Docker-based workload isolation, and a web dashboard to visualize nodes, jobs, and scheduling behavior.
- Implemented 3 pod-placement strategies - First Fit, Best Fit, and Worst Fit - with 30-second heartbeat failure detection and automatic pod rescheduling from failed Docker-backed nodes.

TECHNICAL SKILLS

AI / ML / LLMs: Model Context Protocol (MCP), LLM evaluation, prompt engineering, long-context systems, Claude Desktop/API, TensorFlow, PyTorch, Scikit-learn, Pandas, Librosa

Languages: Python, C++, SQL, Java, JavaScript, HTML/CSS

Backend & Databases: Flask, GraphQL, REST APIs, Spring Boot, Node.js, Express.js, PostgreSQL, Redis, MongoDB, MySQL

DevOps & Tools: Docker, Kafka, Git, VS Code, CI/CD basics, API testing, debugging workflows

Cloud: AWS Lambda, API Gateway, S3, SQS basics

LEADERSHIP

- Teaching Assistant, PES University - Machine Learning: mentored juniors on ML projects, labwork, and assignments.
- Treasurer, Rotaract Prestige Falcon City: managed club funds and budget tracking for student-led initiatives.